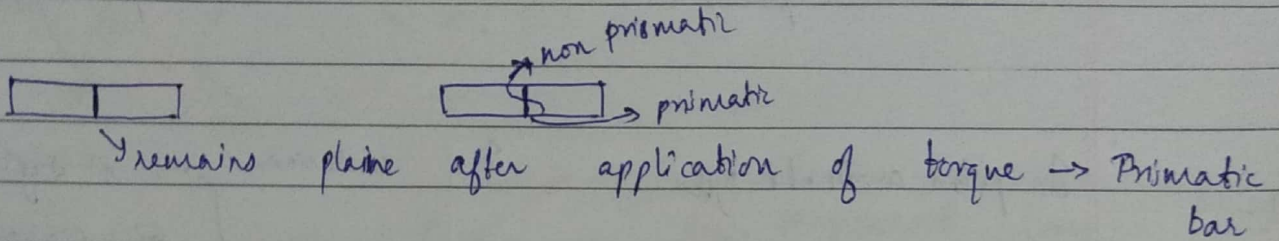
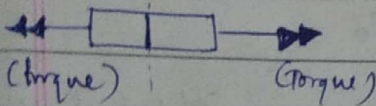


10/11/19
Tuesday

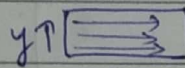
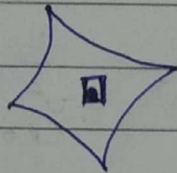
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Membrane → stretched out cloth

→ no stress component normal to the plane. (No normal stress)
(all stress components are within the plane)



→ n (only shear).

- Normal stress

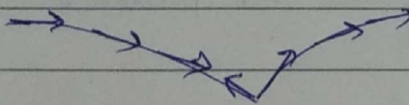
- Shear stress

τ_{xy} or τ_{yx}

τ_{zy}

Plane in which it is acting
(z is normal)

Direction in which stress is acting



Material

Isotropic

pts same along all directions

Homogeneous

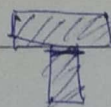
pts same at any 2 pts.

Avg of young's modulus @ a pt
(say)

→ same along all directions

Continuum theory → Theory of elasticity

Only dealing with Linear Deformation

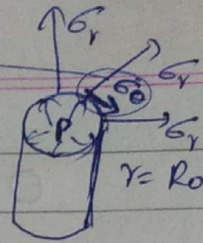


shouldn't be a jump
as it is a continuous

Shear line drawn

Hooke's stress

Internal pressure



cylindrical $\rightarrow \frac{PR}{4T}$

spherical $\rightarrow \frac{PR}{2T}$

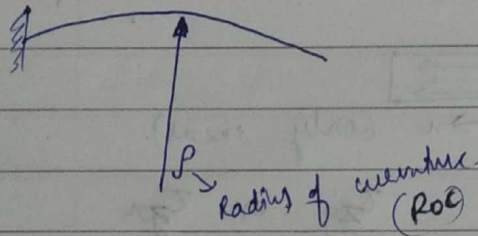
In polar coordinate system

$$\sigma_r = P \text{ at } r = R_o$$

In cartesian

to solve

$$\begin{cases} r^2 + y^2 = R_o^2 \\ \sigma_x = p \cos \theta \\ \sigma_y = p \sin \theta \end{cases}$$



$$K = \frac{1}{\rho} \quad (\text{kappa}) \quad (\text{ROC})$$

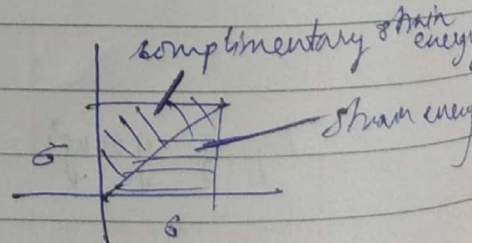
Write cartesian coordinate eqn in polar coordinate

Airy's stress function

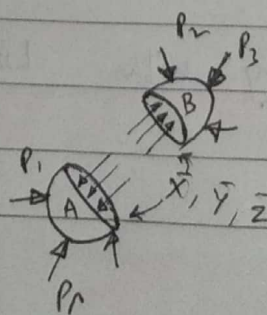
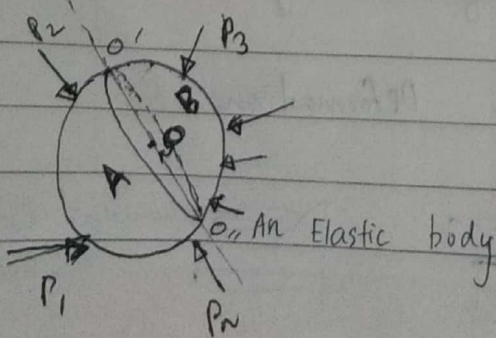
$$(\nabla^2 \phi)^2 = 0$$

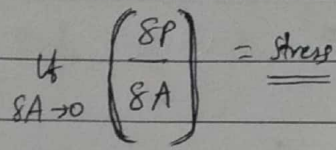
• for 2D problems

Complimentary Strain Energy
Virtual work method

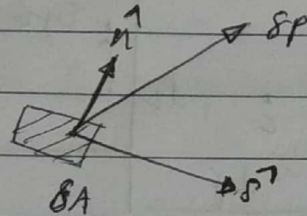


11/1/19
Friday





→ associated with internal forces



$\hat{n} \rightarrow$ normal stress $[\sigma]$

$\hat{s} \rightarrow$ shearing stress $[\tau]$

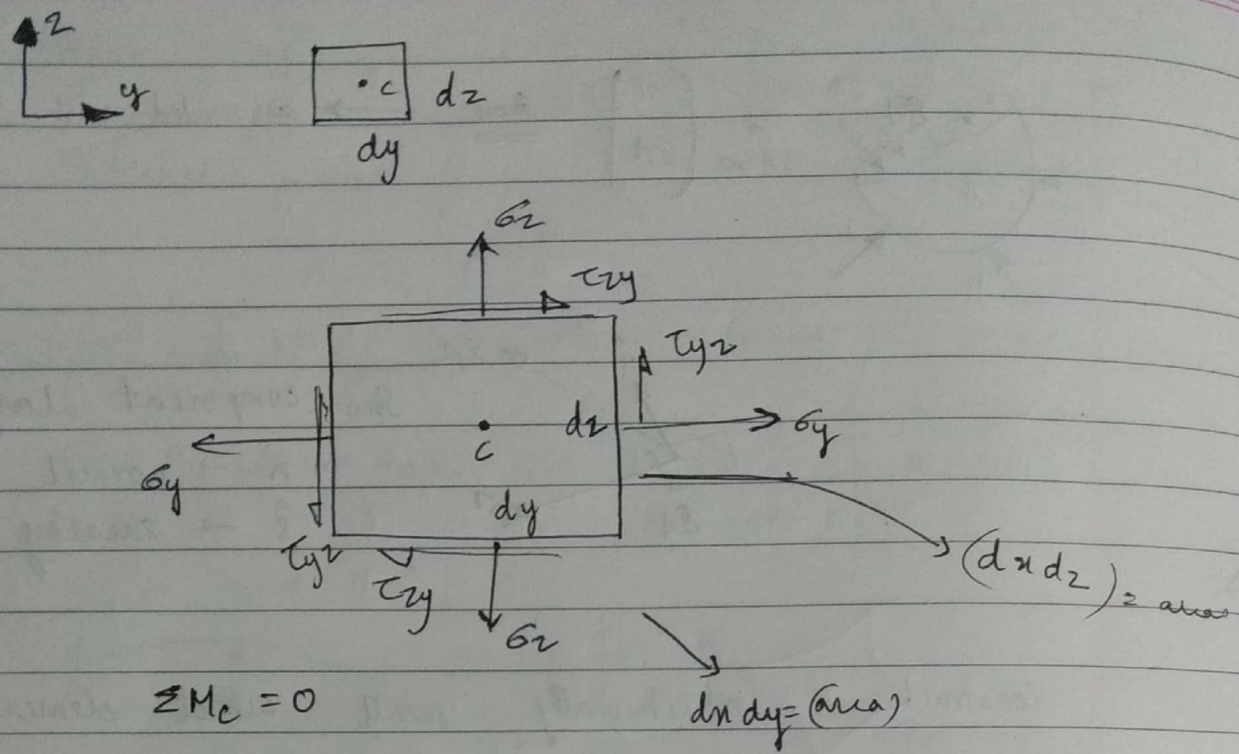
A hand-drawn diagram of a 3D stress element (a cube) in a Cartesian coordinate system (x, y, z). The cube is labeled "Plane" on its top face and "dx dy dz" on its bottom face. The faces are labeled with normal stress components (σ_x , σ_y , σ_z) and shear stress components (τ_{xy} , τ_{yx} , τ_{xz} , τ_{kz} , τ_{zy} , τ_{zx}). The coordinate axes are labeled x, y, and z.

$$dx \, dy \, dz = O(\epsilon)$$

On -ve plane Along -ve y is the } $\Rightarrow$ for E also
On the plane Along the y is the } on -ve plane, direction
of axes changes

Normal stress : $\sigma_x, \sigma_y, \sigma_z$

Shearing stress : $\tau_{xy}, \tau_{xz}, \tau_{yx}, \tau_{yz}, \tau_{zx}, \tau_{zy}$ } 9 components of stresses

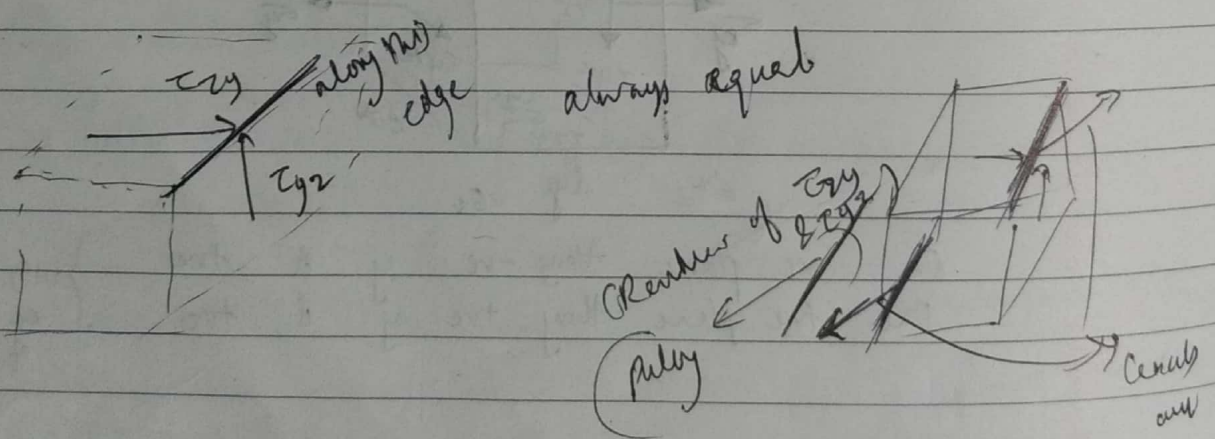


$$\sum M_c = 0$$

Taking moments

$$(\tau_{yz})(dx \, dz) \cdot dy - \tau_{zy} \, dn \, dy \cdot dz = 0$$

$$\tau_{yz} = \tau_{zy}$$



Taking y plane

$$\tau_{zx} = \tau_{xz}$$

Taking z plane

$$\tau_{yx} = \tau_{xy}$$

$\sigma_x \sigma_y \sigma_z \quad \tau_{xz} \tau_{xy} \tau_{zy}$

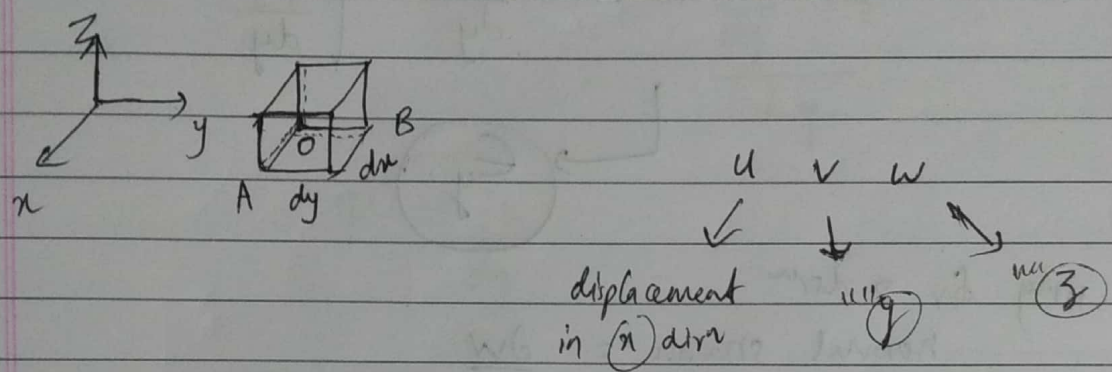
Eulerian
 ↓
 control volume
 ↓
 passing through the given volume

(Lagrangian) we are in the only one pt

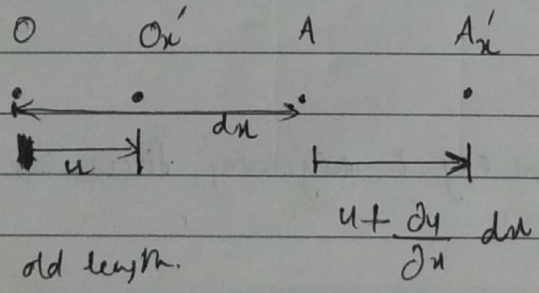
9 - 3 = 6 components are required to define state of stress @ a pt (as 3 are equal among 9)

State of strain @ a pt

Strain & components of strain (at a pt)



After force applied, O moves to Ox'
 A moves to Ax'



$dx \rightarrow$ old length
 $dx' \rightarrow$ new length

$$dx' = \left(u + \frac{\partial u}{\partial x} dx \right) - u + dx$$

change in length $\delta x = \frac{\partial u}{\partial x} \cdot dx$

Normal strain / unit change in length = $\frac{\partial u}{\partial x} \cdot \frac{dx}{dx}$

$\rightarrow \boxed{\epsilon_x}$
 (epsilon)

or $\frac{\delta x}{dx}$

$$\tan \phi \approx \frac{\frac{\partial v}{\partial x} \cdot dx}{dx}$$

for small angle

$$\phi \approx \frac{\partial v}{\partial x}$$

IIIly

$$\tan \psi \approx \frac{\frac{\partial u}{\partial y} \cdot dy}{dy}$$

$$\psi \approx \frac{\partial u}{\partial y}$$

change in angle = $\phi + \psi = \boxed{\gamma_{xy} = \frac{\partial u}{\partial y} + \frac{\partial v}{\partial x}}$
 [b/w two planes which were originally $\perp$ r.]
 shear strain in the xy plane.

IIIly

in the xz plane

$$\boxed{\gamma_{yz} = \frac{\partial v}{\partial z} + \frac{\partial w}{\partial y}}$$

$$\boxed{\gamma_{xz} = \frac{\partial u}{\partial z} + \frac{\partial w}{\partial x}}$$

in the yz plane